

E9 205 Machine Learning for Signal Processing

**Supervised-Dimensionality-Reduction.
Decision Theory
Probability Distributions**

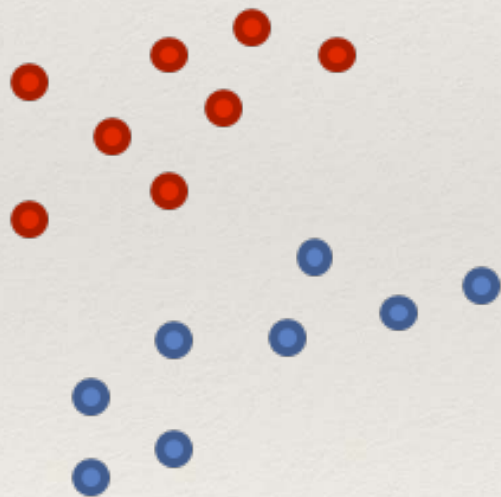
26-08-2019

Advantages and Disadvantages of PCA

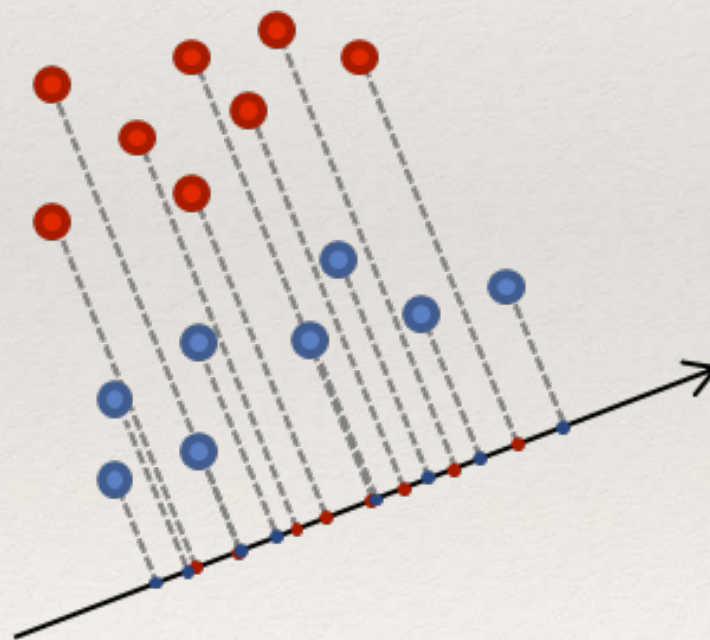
- ❖ Simple linear transform
 - ❖ Eigen decomposition of Data Covariance matrix is straight forward.
 - ❖ PCA for high dimensional data ?
- ❖ Variance maximization may not be the ideal loss function in dimensionality reduction.
- ❖ If the data contains discrete class labels, we can do better than PCA to maximize class separation.

Need for Supervised Dimensionality Reduction

Labelled
data

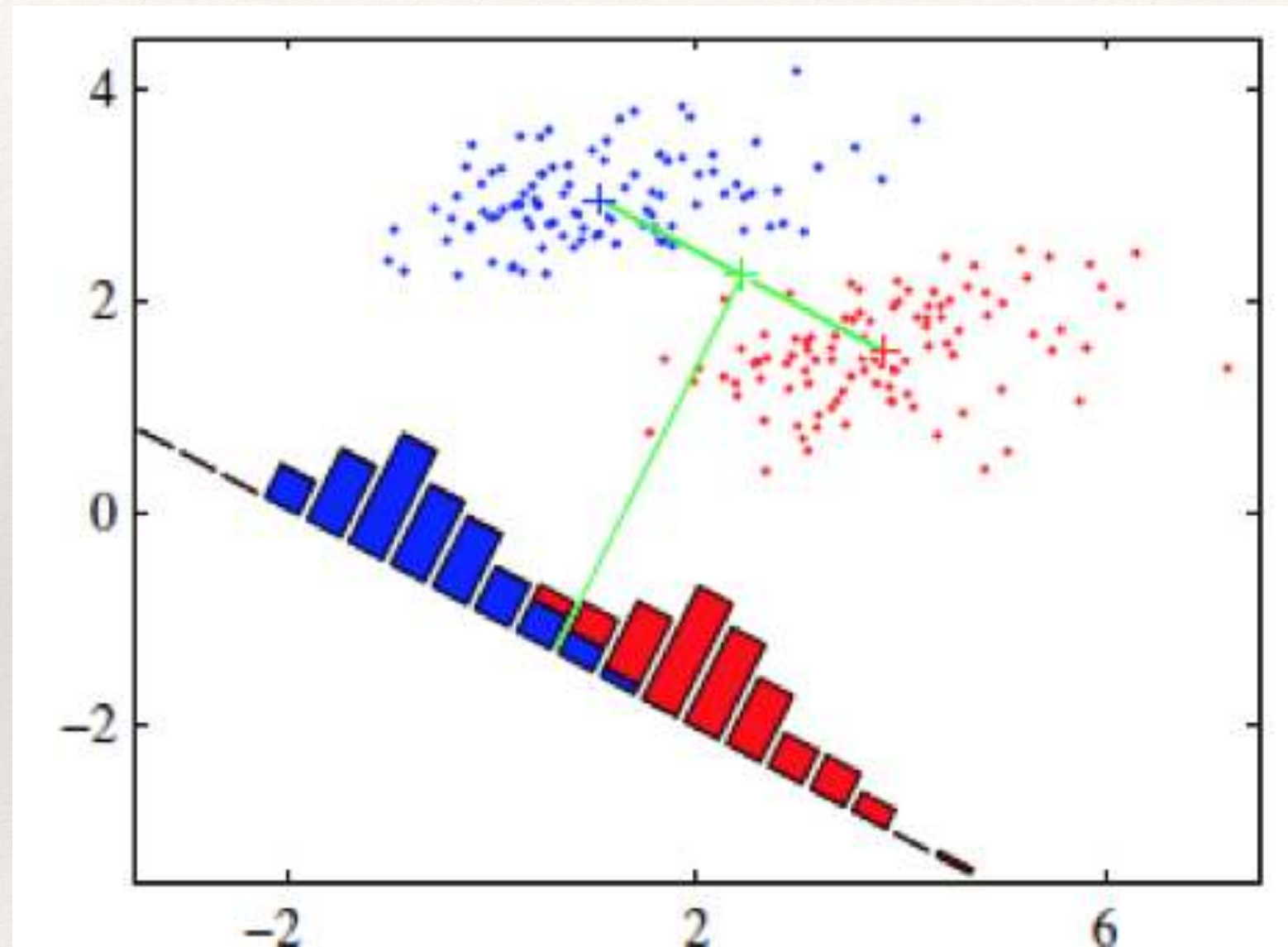


PCA projection:
Maximising the variance of
the whole set



Linear Discriminant Analysis

Without the Within Class Factor



Linear Discriminant Analysis

Find a linear transform $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$ with a criterion which maximizes the class separation

- Maximize the between class distance in the projected space while minimizing the within class covariance

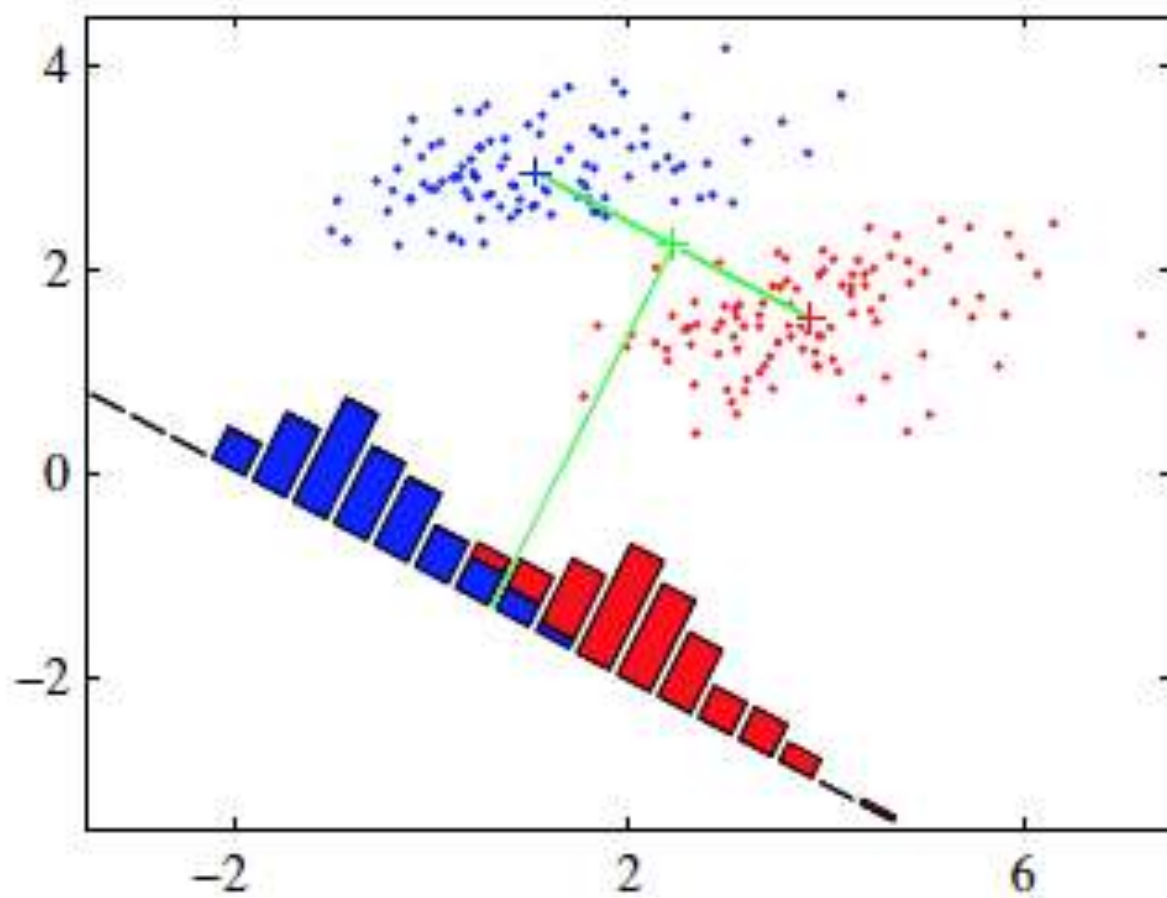
$$J = \frac{\mathbf{w}^T \mathbf{S}_b \mathbf{w}}{\mathbf{w}^T \mathbf{S}_w \mathbf{w}}$$

$$\mathbf{S}_b = \sum_{k=1}^K N_k (\mathbf{m}_k - \mathbf{m})(\mathbf{m}_k - \mathbf{m})^T \quad \mathbf{S}_w = \sum_{k=1}^K \sum_{n \in C_k} (\mathbf{x}_n - \mathbf{m}_k)(\mathbf{x}_n - \mathbf{m}_k)^T$$

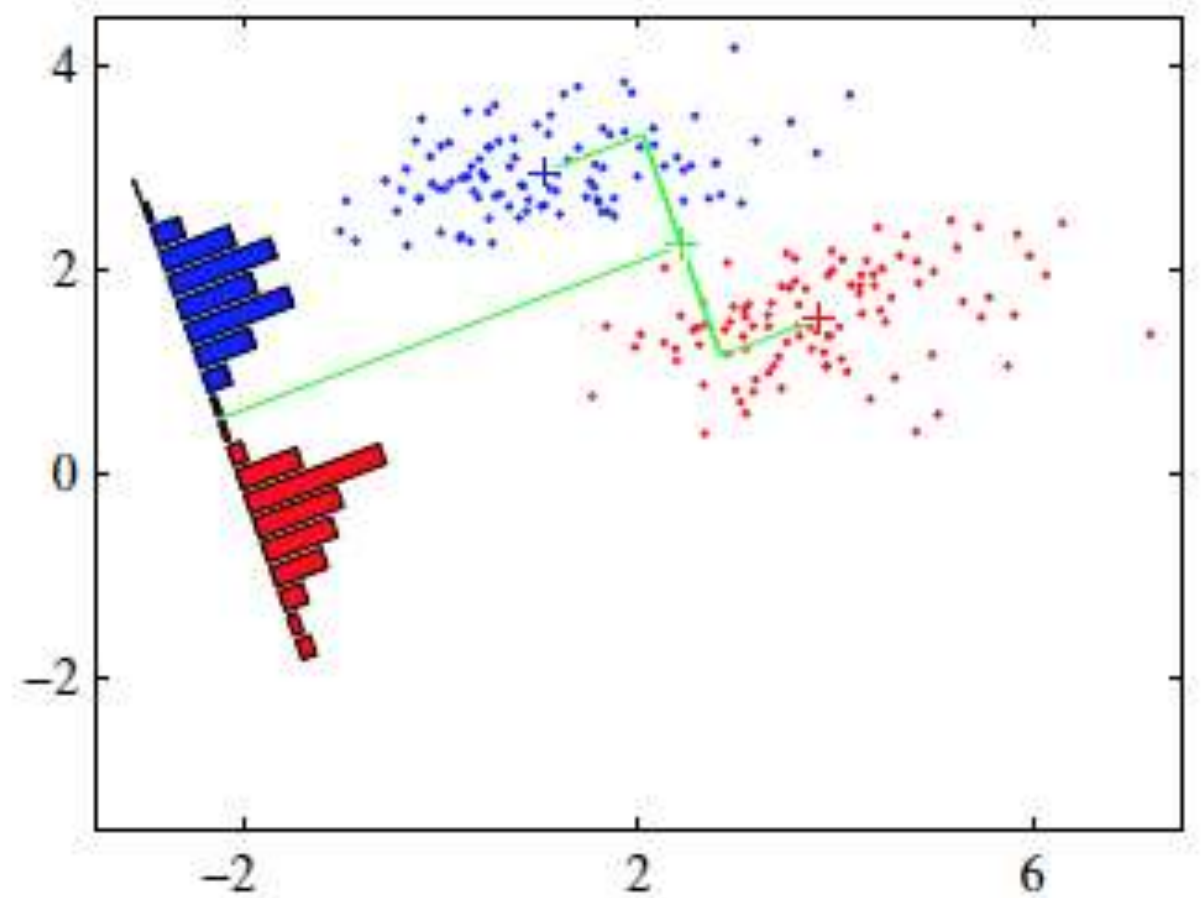
- ❖ Generalized Eigenvalue problem
- ❖ Eigenvectors of $\mathbf{S}_w^{-1} \mathbf{S}_b$

Linear Discriminant Analysis

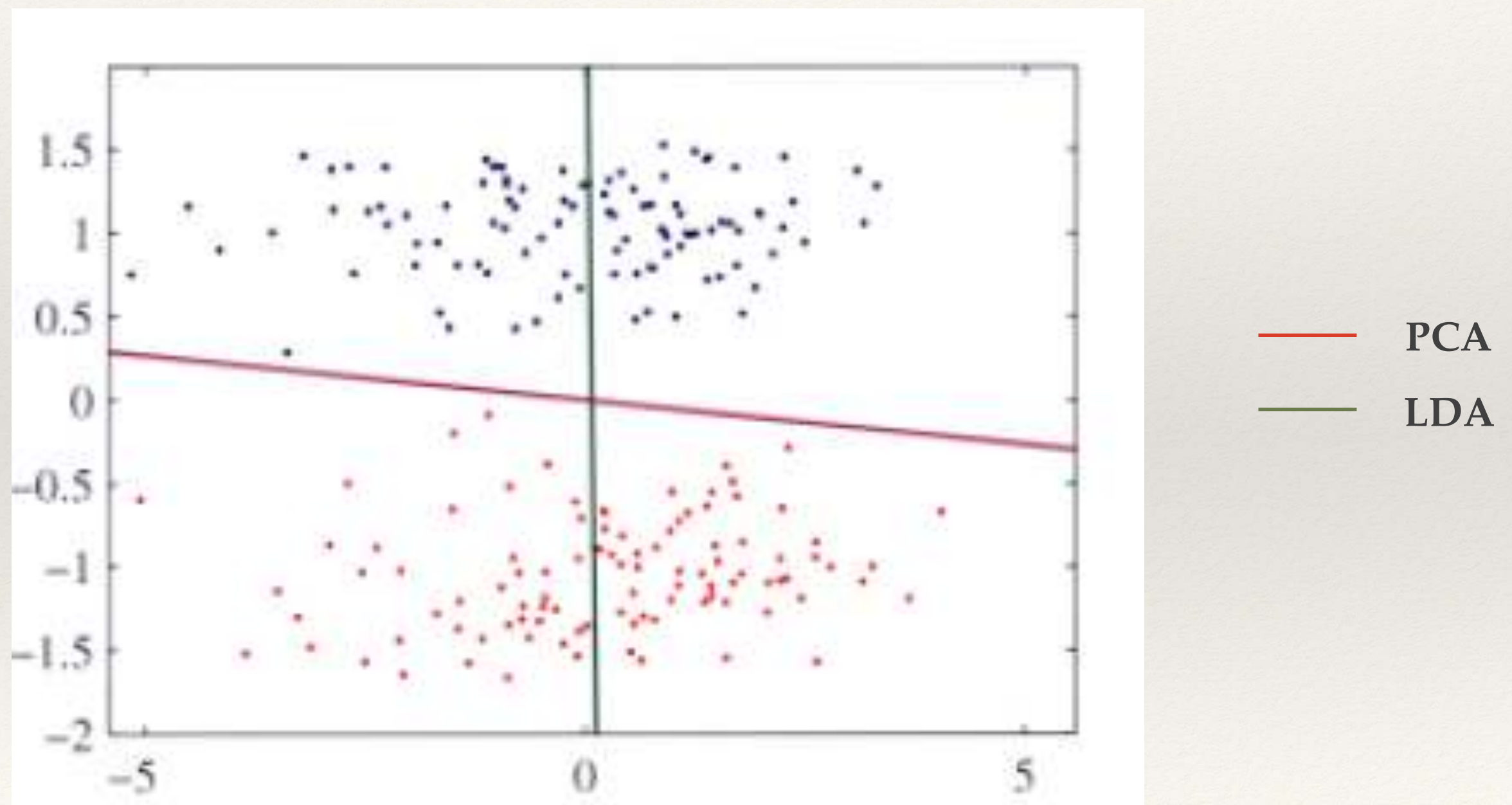
Projecting on line joining means



Fisher Discriminant

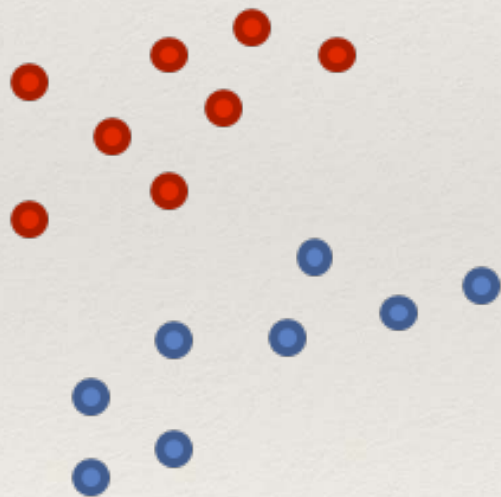


PCA versus LDA

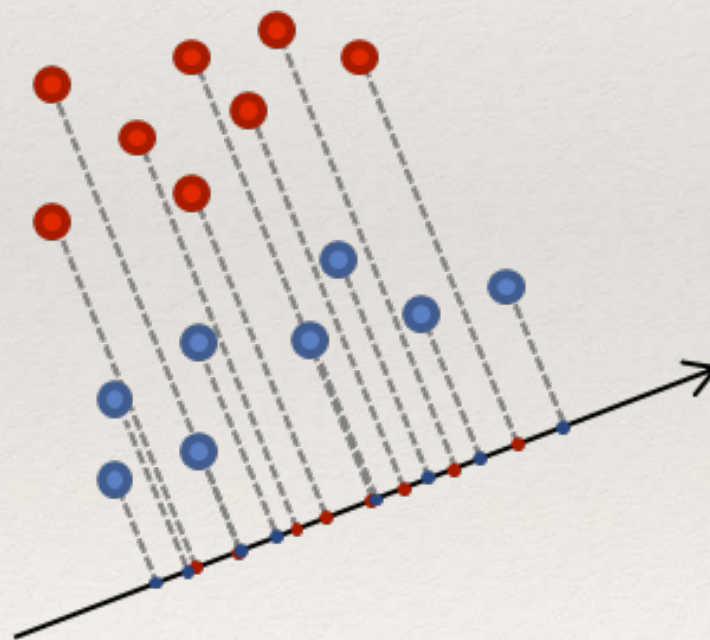


PCA versus LDA

Labelled
data



PCA projection:
Maximising the variance of
the whole set



LDA projection:
Maximising the distance
between groups

